

John Paul Nyunja

Senior Full-Stack Engineer | Systems Architect | AI Automation Specialist

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PROFESSIONAL SUMMARY

Senior Full-Stack Engineer with 5+ years of experience specializing in building high-performance, accessible, and scalable React ecosystems. Expert in modern frontend architecture using **React 19, Next.js 15, TanStack Router, and TanStack Query**. Proven track record of collaborating with cross-functional teams to deliver pixel-perfect, accessible (WCAG-compliant) interfaces that bridge complex backend data with intuitive user experiences. I also specialize in **Go, TypeScript, and Python**, focusing on high-performance applications that balance low-level efficiency with a smooth user experience. My background includes extensive work with fintech APIs, where I've managed over 1M+ monthly calls and successfully reduced MTTR for Tier 1 issues to under 24 hours. Passionate about leveraging clean code, type-safe development (TypeScript), and automated testing to build robust, user-centric web applications. By combining a deep understanding of Linux internals and algorithmic optimization with modern frameworks like React 19 and Next.js 15, I build scalable, end-to-end solutions that solve real-world problems.

CORE TECHNICAL COMPETENCIES

Frontend & Mobile Development

- **Frameworks:** React 19, Next.js 15 (App Router), React Native, Vite.
- **State & UI:** TanStack Query, Redux, Tailwind CSS, Shadcn/UI, Framer Motion, Recharts.
- **Tooling:** Playwright (E2E), Jest, Storybook, GraphQL Apollo.

Backend & Systems Engineering

- **Languages:** Go (Golang), Node.js, TypeScript, Python, Rust, C, SQL, R.
- **Architecture:** Microservices, Clean Architecture, Distributed Systems, RESTful APIs, gRPC, GraphQL, WebSockets.
- **Low-Level:** Custom RDBMS development, B-Tree Indexing, Storage Engines, Linux/Unix Internals, Memory Management.
- **Database:** PostgreSQL (sqlc/GORM), Supabase, Redis (Caching/Pub-Sub), SQLite, MongoDB.

AI, Machine Learning & Automation

- **Generative AI:** Gemini Pro API, OpenAI SDK, LLM-based autonomous agents, Prompt Engineering.

- **Data Science:** C4.5 Decision Tree implementation, Genomic data pipelines, Web Scraping (Go/Python).
- **Automation:** Browser automation via Playwright/Puppeteer, GitHub Actions CI/CD, Cron Job scheduling.

Cloud, DevOps & Infrastructure

- **Orchestration:** Docker, Kubernetes (KCNA Certified).
- **CI/CD:** Jenkins, GitHub Actions, Vercel, Render.
- **Monitoring:** Prometheus, Grafana, Structured JSON Logging (Zap/Slog).
- **Security:** JWT/OAuth2, Idempotency keys, Rate Limiting, Webhook Verification.

PROFESSIONAL EXPERIENCE

Nexveria Solutions | *Full-Stack Software Developer*

January 2024 – Present

- **Architected and deployed** an enterprise-grade Go/PostgreSQL backend for a multi-tenant property management ecosystem, supporting 500+ tenants with automated rent collection.
- **Integrated Twilio API** for automated payout notifications and SMS-based lease reminders, increasing tenant response rates by 40%.
- **Optimized data persistence layer** using sqlc and Redis caching, resulting in a 60% reduction in API response times for heavy financial reporting queries.
- **Engineered webhook retry logic** with exponential backoff, ensuring 98.5% delivery success for payment confirmations in low-bandwidth environments.
- **Developed high-performance UI** using Next.js 15 and Shadcn/UI, featuring real-time financial dashboards and secure document management.
- Implemented complex state management and data synchronization using **TanStack Query**, reducing latency and ensuring a seamless, real-time interface for 500+ tenants.
- Collaborated directly with designers and backend engineers to translate wireframes into accessible, interactive components, ensuring full **WCAG compliance and keyboard accessibility**.
- **Engineered 5+ Go-based REST APIs** handling 1M+ monthly requests with a focus on concurrency and zero-downtime deployments.
- **Implemented custom graph-based pathfinding algorithms** in Go to solve complex network flow challenges, achieving 100% efficiency in non-overlapping path identification.
- **Mentored a cohort of 20+ developers** on Go concurrency patterns (goroutines/channels) and clean architecture principles.
- **Automated deployment pipelines** using GitHub Actions and Docker, reducing manual deployment errors by 35%.

Syncfusion | *Technical Support Engineer*

2023 – 2024

- **Diagnosed and resolved complex API failures** for 1,000+ enterprise accounts, maintaining a 90% first-contact resolution rate and meeting 24-hour MTTR targets.
- **Debugged JWT authentication flows** and SQL query bottlenecks in partner integrations, cutting latency-related complaints by 40%.
- **Authored production-grade documentation** and SDK troubleshooting guides, reducing Tier 1 support escalations by 25%.
- **Created "unbreakable" sample applications** in Go and Node.js to assist enterprise clients in debugging CI/CD pipeline integrations.

Freelance Consultant | *Full-Stack & Systems Engineer*

2021 – 2023

- **Developed and maintained** custom payment gateways for 5+ African fintech startups, resolving over 200 critical edge cases in transaction processing.
- **Built a custom RDBMS from scratch** in Go to manage micro-payment ledgers, implementing manual file I/O and B-Tree indexing to demonstrate deep systems proficiency.
- **Designed automated data scrapers** in Go and Python to aggregate financial and real estate data, processing 50k+ records into structured formats for market analysis.
- **Deployed AI-powered autonomous agents** for recruitment automation, utilizing LLMs to perform intelligent form filling and job application tracking.

JOOUST | *Research Data Analyst*

2021 – 2022

- **Automated genomic data pipelines** for COVID-19 research, utilizing Go and PostgreSQL to process 300+ participant datasets, reducing manual data entry time by 30%.
- **Developed API-like endpoints** for real-time academic reporting and data validation.

SELECTED TECHNICAL PROJECTS

Rentbase | *Full-Stack Property Management SaaS*

- **Role:** Lead Architect
- **Tech:** Go, Next.js 15, PostgreSQL, Docker, TypeScript, Tailwind CSS.
- **Impact:** Engineered a secure, multi-tenant architecture that streamlines property oversight. Features include automated lease management, maintenance request tracking via WebSockets, and real-time financial reporting.
- **Key Challenge:** Implementing a robust permission system for various user roles (Landlord, Tenant, Manager) within a single unified API.
- **Repository:** github.com/nyunja/rentbase

Simple RDBMS | *Low-Level Database Engine*

- **Role:** Systems Engineer

- **Tech:** Go, Binary File I/O, Algorithms, Data Structures.
- **Impact:** Built a relational database management system from the ground up to handle high-frequency micro-payment ledgers. Implemented custom indexing and ACID-compliant transaction logic without external DB dependencies.
- **Key Challenge:** Managing page-level storage and memory mapping to ensure data persistence and query efficiency.
- **Repository:** github.com/nyunja/simple-rmbs

Fity Budget | *AI-Powered Financial Orchestrator*

- **Role:** Full-Stack Developer / AI Engineer
- **Tech:** Go (Gin), React 19, Gemini AI, GORM, PostgreSQL, Recharts.
- **Impact:** Revolutionized personal finance tracking by integrating Generative AI to categorize expenses and provide predictive financial insights. Reduced manual user input by 60% through automated receipt parsing and NLP.
- **Live Site:** fity-budget.vercel.app

Ghost Apply | *Autonomous AI Job Agent*

- **Role:** Lead Developer
- **Tech:** JavaScript, Playwright, LLM (OpenAI/Gemini).
- **Impact:** Developed an autonomous agent that navigates job boards, interprets complex form fields using LLMs, and submits applications. Features a dashboard to track agent status and success rates.
- **Repository:** github.com/nyunja/ghost-apply

Digital Ant Farm V2 | *Algorithmic Flow Optimizer*

- **Role:** Backend Engineer
- **Tech:** Go, Graph Theory, Edmonds-Karp/BFS.
- **Impact:** Solved high-complexity pathfinding problems by identifying maximum non-overlapping paths through a network of nodes. Optimized for speed and memory, handling graphs with thousands of vertices in milliseconds.
- **Repository:** github.com/nyunja/digital-ant-farm-v2

Social Network | *Real-Time Communication Platform*

- **Role:** Backend Developer
- **Tech:** Go, Next.js, SQLite, WebSockets.
- **Impact:** Created a full-featured social platform including real-time chat, notifications, and feed updates. Leveraged Go's concurrency model to manage thousands of simultaneous WebSocket connections.
- **Repository:** github.com/tajjjjr/social-network

ADDITIONAL OPEN SOURCE & RESEARCH PROJECTS

- **KBase Bulk Uploader:** Python/Playwright tool for automated genomic data ingestion into KBase Narratives.
- **C4.5 Decision Tree:** A high-performance, scalable implementation of the C4.5 machine learning algorithm in Go for classification tasks.
- **Wallet Transaction API:** A production-ready REST API for digital wallet management, featuring strict idempotency and transaction locking.
- **NotezJiji:** An e-commerce marketplace for academic materials built with React and Supabase, featuring secure payment processing.
- **Webby Scraper:** A Go-based concurrent web scraper for vehicle price aggregation and market trend analysis.
- **Therapy Chatbot:** An LLM-powered simulation chatbot using Gemini-Pro to provide text-based therapeutic conversation.

EDUCATION

Bachelor of Science (BSc) | Kenyatta University

Graduated 2015

Self-Taught Advanced Software Engineering

2021 – Present

- Extensive focus on distributed systems, compiler design, and AI integration.

CERTIFICATIONS

- **KCNA (Kubernetes and Cloud Native Associate)** – Cloud Native Computing Foundation (CNCF)

SOFT SKILLS & METHODOLOGIES

- **Leadership:** Mentorship of junior developers, technical lead on cross-functional teams.
- **Analytical Thinking:** Root cause analysis (RCA), bug triaging, and performance bottleneck identification.
- **Communication:** Technical writing (OpenAPI/Swagger), stakeholder management, and solution architecture presentations.
- **Methodologies:** Agile/Scrum, Test-Driven Development (TDD), CI/CD, Domain-Driven Design (DDD).